

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Mechanical Engineering Department

2. Subject Name: Casting, Forming and Welding L-T-P: 3-1-0 Credits:4

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

Casting: Introduction to sand casting process, cast metals and alloys. Pattern design, Green sand moulding and sand testing, Core, Chaplet and Chill. Microstructure in casting, dendrite structure and CFR, Heat transfer and solidification time (Chvorinov's rule). Feeder design. Gating design, Fluidity. Different other casting processes: Die casting, Squeeze casting, Centrifugal casting, Investment casting, Continuous casting, Casting defects. Forming: Introduction to forming processes, basic structure of metal and heat treatments, Plastic deformation and Yield criteria. Different conventional forming processes: rolling, forging, extrusion, drawing, deep drawing. Slab analysis for force estimation, forming dies and tools, forming defects. Different nonconventional forming processes: Electromagnetic, hydraulic and explosive forming, Powder metallurgy: Applications, methods, advantages and disadvantages. Welding: Introduction to welding processes, Welded Joint types, Weld specifications, Symbols, Conventional Fusion Welding Processes: Gas welding, Shielded metal arc welding (SMAW), Gas metal arc welding (GMAW), Submerged Arc welding, Gas Tungsten Arc Welding (GTAW or TIG), Physics of Arc welding, Metal Transfer in Arc Welding, Power sources, Heat flow in welds, The welding thermal cycle, Time-Temperature curves, The generalized heat flow equation. Other welding processes: Solid-state welding processes, High-Energy-Density Beam Welding Processes. Design of Weld joints (simple joints), Weld Defects & Discontinuities. Weld Quality & Destructive and Non-destructive Testing.

Reference Books:

- 1) Casting: (a) Manufacturing Technology (Vol 1) by P N Rao; (b) Complete Casting Handbook by J Campbell
- 2) Forming: (a) Manufacturing Science by Amitabha Ghosh and Asok Kumar Mallik; (b) Principles of Industrial Metalworking Processes by G W Rowe
- 3) Welding: (a) Principles of Welding by Robert W Messler; (b) Manufacturing Engineering and Technology by S Kalpakjian

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Casting: Introduction to sand casting process, cast metals and alloys. Pattern design, Green sand moulding and sand testing,	3
2.	Core, Chaplet and Chill. Microstructure in casting, dendrite structure and CFR, Heat transfer and solidification time (Chvorinov's rule).	4
3.	Feeder design. Gating design, Fluidity.	4
4.	Different other casting processes: Die casting, Squeeze casting, Centrifugal casting, Investment casting, Continuous casting, Casting defects	3
5.	Forming: Introduction to forming processes, basic structure of metal and heat treatments, Plastic deformation and Yield criteria.	3
6.	Different conventional forming processes: rolling, forging, extrusion, drawing, deep drawing.	2
7.	Slab analysis for force estimation, forming dies and tools, forming defects.	6
8.	Different nonconventional forming processes: Electromagnetic, hydraulic and explosive forming, Powder metallurgy: Applications, methods, advantages and disadvantages.	3
9.	Welding: Introduction to welding processes, Welded Joint types, Weld specifications, Symbols,	3
10.	Conventional Fusion Welding Processes: Gas welding, Shielded metal arc welding (SMAW), Gas metal arc welding (GMAW), Submerged Arc welding, Gas Tungsten Arc Welding (GTAW or TIG),	3
11.	Physics of Arc welding, Metal Transfer in Arc Welding, Power sources, Heat flow in welds, The welding thermal cycle, Time-Temperature curves, The generalized heat flow equation.	4
12.	Other welding processes: Solid-state welding processes, High-Energy-Density Beam Welding Processes. Design of Weld joints (simple joints), Weld Defects & Discontinuities. Weld Quality & Destructive and Non-destructive Testing.	4
13.	Tutorial on casting, forming and welding	6
14.	Class test-1 and Class test-2	2
Total number of hours		50